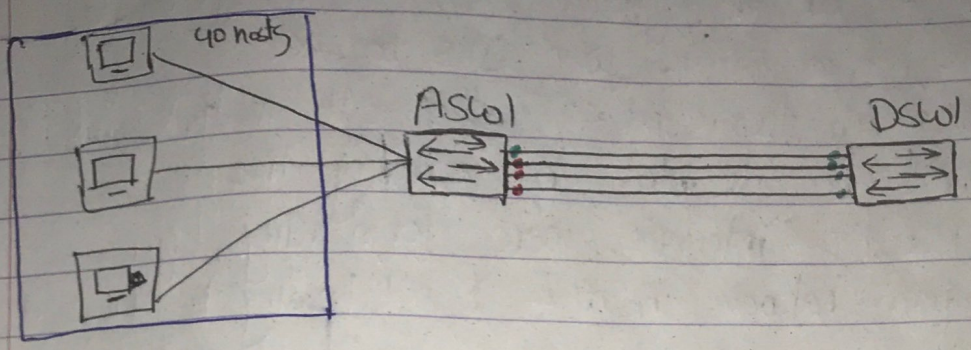
Day-23 Dec-17

Ether-Channel.



ASw1 = Access layer switch, a switch that end hosts connect to
DSw1 = Distribution layer switch, a switch that access layer switches connect to

All the 40 end hosts are trying to access to internet to do their work. The network admin notices that connection to DSw1 is congested, so he decides to add other links to increase the bandwidth so that it can support all of the end hosts.

Adding the 3 links doesn't help & connection between ASw1 & DSw1 is still congested.

NOB: When the bandwidth of the interfaces connected to end hosts is greater than the bandwidth of the connection, to the distribution switch (es) this is called oversubscription.
Some oversubscription is acceptable, but too much will get cause congestion,

The congestion doesn't seem to improve, so network admin adds another link. Now there are 4 links between ASw1 & DSw1.

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Now, will the congestion decreases / be better?

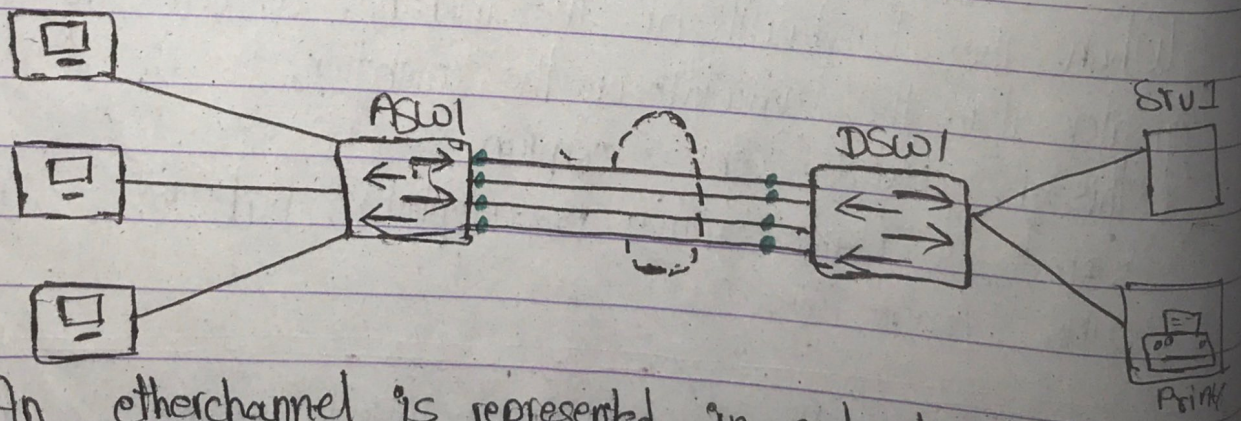
↳ When observed all link-lights of DSW1 are green but only one is green in ASW1. It's because:-

- if we connect all two switches together with multiple links, all except one'll be disabled by spanning tree.
- If all of ASW1's interfaces were forwarding, layer 2 loops would form between ASW1 & DSW1, leading to broadcast storms.
- Other links will be unused unless the active link fails. In that case, one of the inactive links'll start forwarding.

Although having 4 links provide redundancy, its waste of bandwidth as 3 of them are inactive.

- If we group these 4 physical interface as one logical interface, Etherchannel can solve this problem by giving us both redundancy & increased bandwidth.

Etherchannel.



- An etherchannel is represented in network diagrams by drawing a circle around the interfaces.

- EtherChannel groups multiple interfaces together to act as a single interface.
- STP will treat this group as a single interface.
  Now all 4 links'll have green light as STP treat it as 1 link.
- Traffic using the EtherChannel will be load balanced among the physical interfaces in the group. An algorithm is used to determine which traffic will use which physical interface.
- If the four links were each of 1 gigabit, now it acts as a virtual 4-gigabit interface.
- Some other names for etherchannel are:-
  Port Channel
  LAG (Link Aggregation Group)

EtherChannel load balancing.

Ether channel load balances based on flows

- A flow is a communication between two nodes in the network.
- Frames in the same flow will be forwarded using the same physical interface.
- If frames in same flow were forwarded using different physical interfaces, some frames may arrive at the destination out of order, which can cause problems

- The calculation that is done to determine which physical interface to use takes into account a few inputs.
  We can change the inputs used in interface selection calculation.

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Inputs that can can be used:
a) Source MAC - source with same MAC will always use the same interface in the ether channel.
b) Destination MAC
c) Source & Destination MAC
d) Source IP
e) Destination IP
f) Source & Destination IP

Configuring Ether-Channel Load-Balancing

1) ASW1 # show etherchannel load-balance
src-dst-ip
2) ASW1 (config)# port-channel load-balance src-dst-mac
3) ASW1 (config)# do sh etherchannel load-balance
src-dst-mac

NOB: To check load-balance status use "etherchannel" to configure load-balance use "port-channel"

EtherChannel Configuration

There are three methods of EtherChannel Configuration on CISCO switches:

A) PAgP (Port Aggregation Protocol)
↳ Cisco proprietary protocol
↳ Dynamically negotiates the creation/maintenance of the EtherChannel. (like DTP does for trunks)

B) LACP (Link Aggregation Control Protocol) (V.I. for CCNA Syllabus 2nd attempt, but study all)

↳ Industry Standard (IEEE 802.3ad).
↳ Dynamically negotiates the creation/maintenance of the EtherChannel (like DTP does for trunks).

C) Static EtherChannel.
↳ A protocol is not used to determine if an EtherChannel should be formed.
↳ Interfaces are statistically configured to form an Ether Channel

- Upto 8 interfaces can be formed into single EtherChannel (LACP allows upto 16, but only 8 will be active, the other 8 will be in standby mode, waiting for an active interface to fail).

PAgP configuration.

```
ASW1(config)# int range g0/0-3
ASw1(config-if)# channel-group 1 mode ?
```

active	Enable LACP unconditionally.
auto	Enable PAgP only if a PaGP device is detected
desirable	Enable PaGP unconditionally
on	Enable Etherchannel only
passive	Enable LACP only if LACP device is detected

auto + auto = no EtherChannel
desirable + auto = EtherChannel
desirable + desirable = EtherChannel

ASW1(config-if-range) # channel-group 1 mode desirable

Command
SW1# channel-group # number mode prefe_mode

NOB:

The channel-group number has to match for member interfaces on the same switch. However it doesn't have to match the channel group number on another switch.

LACP configuration.

instead of desirable → active
instead of auto → passive mode
Other things are same PAGP protocol,

passive + passive = no EtherChannel.
active + passive = Ether Channel
active + active = Ether Channel.

eg SW1 # channel-group 1 mode active

Static Ether Channel Configuration.

There aren't two separate modes, just one 'ON' which manually tells these interfaces to form Ether Channel.

ASW1# channel-group 1 mode on.

On mode only works with on mode
(on + desirable or on + active won't work).

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Manually configure the Negotiation Protocol

```
ASW1# channel-protocol lacp
ASW1 # channel-protocol pagp
```

EtherChannel configuration.

Once we have made etherchannel, we can configure it same way as interface.

```
ASW1# in r interface port-channel 1
        # switchport trunk encapsulation dot1q
        # switchport mode trunk.
```

The settings we apply on port-channel interface will also apply on physical interfaces.

- Member interfaces must have matching configurations
  - ↳ Same duplex (full/half)
  - ↳ Same speed
  - ↳ Same switchport mode (access/trunk)
  - ↳ Same allowed VLAN/native VLAN (for trunk interface)

Some useful commands./notes.

1) show etherchannel summary
2) Even if we're using EtherChannel, layer 2 loops can still occur if multiple switches are connected together in loop

Layer 3 etherchannel

ASW1 # int range g0/0-3
ASW1 (config-if-range) # no switchport
ASW1 (config-if-range) # channel-group 1 mode active.
Creating a port-channel int port-channel

Commands

1) SW(config) port-channel load-balance mode.
configures the Ether-Channel load-balancing method on switch

2) show etherchannel load-balance.
displays information about the load-balancing settings

3) channel-group number mode {desirable | auto | active | passive | on}
configures an interface to be part of EtherChannel

4) show etherchannel summary.
displays a summary of EtherChannels on the switch.

5) ~~show ether chann-~~.

5) show etherchannel port-channel
displays info about the virtual port-channel interface on switch

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